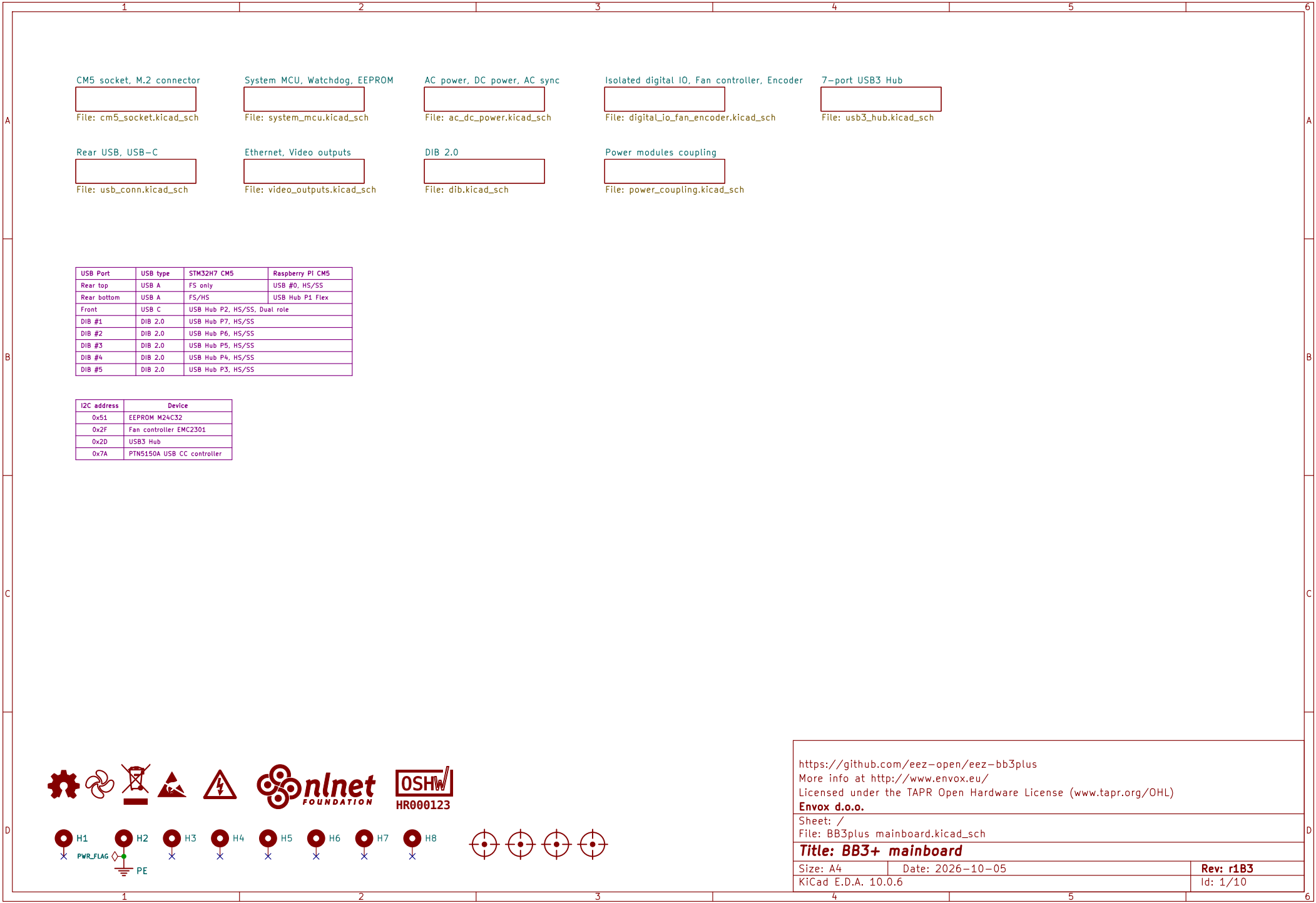
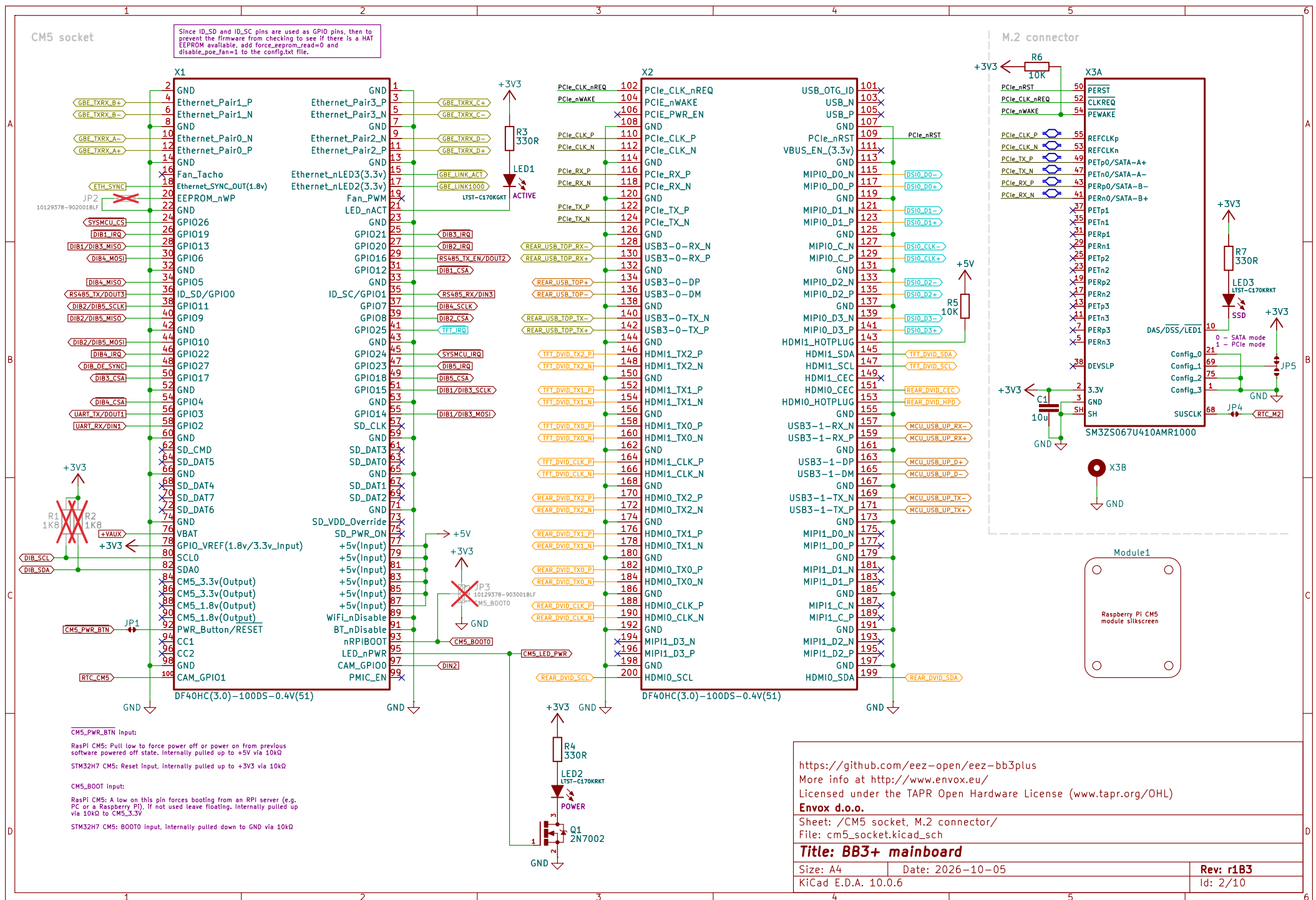
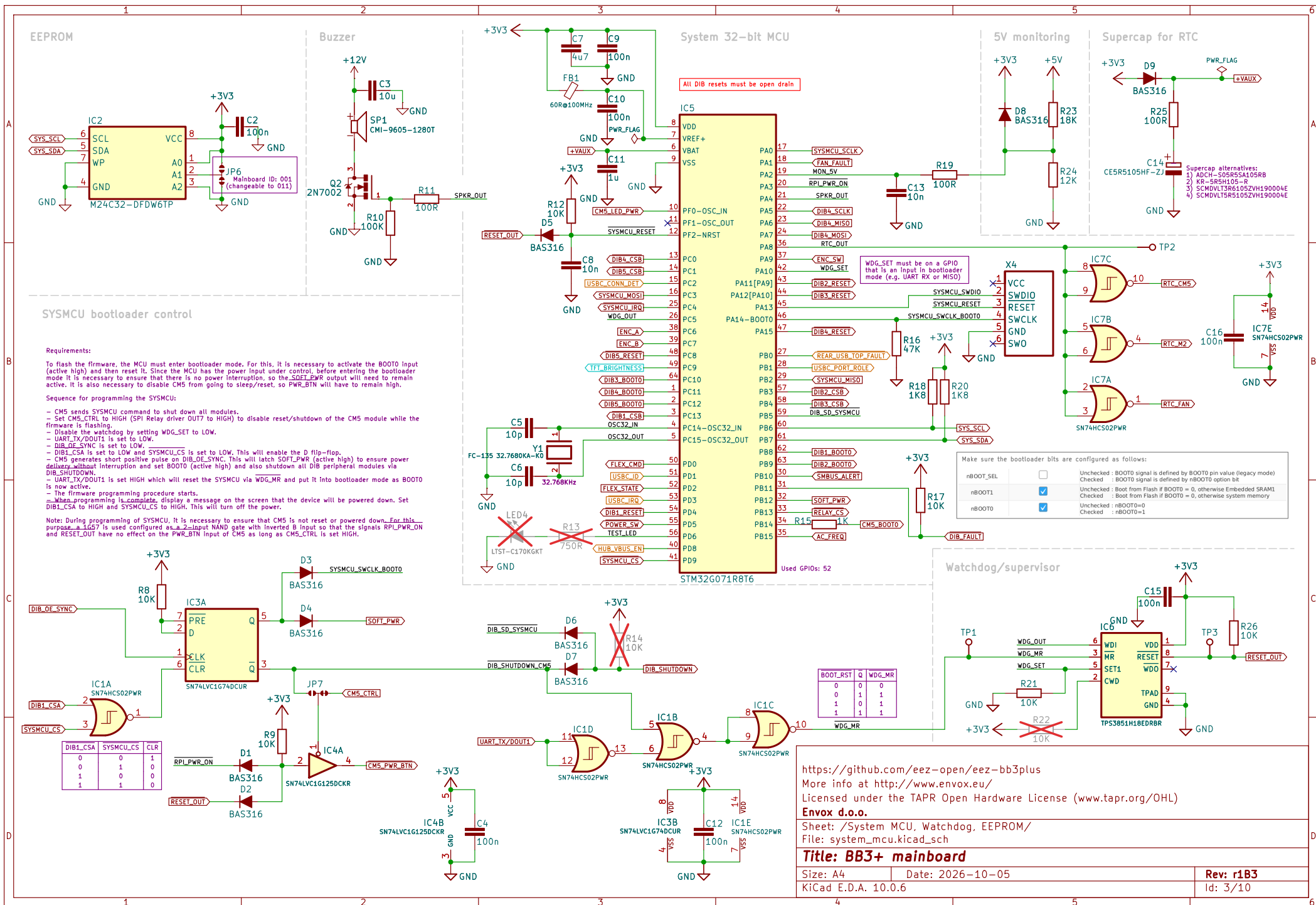
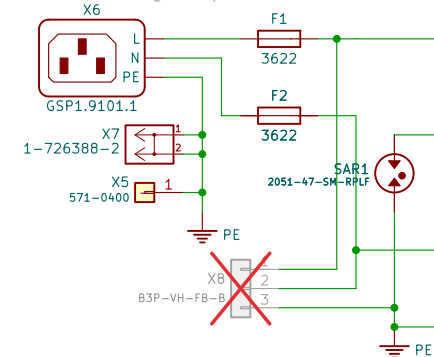




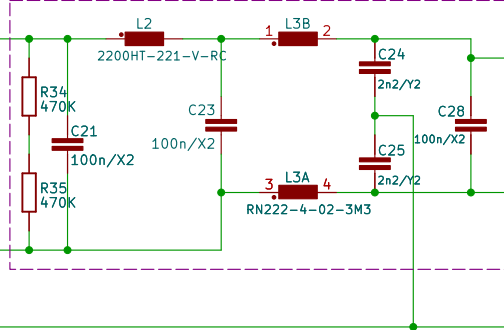




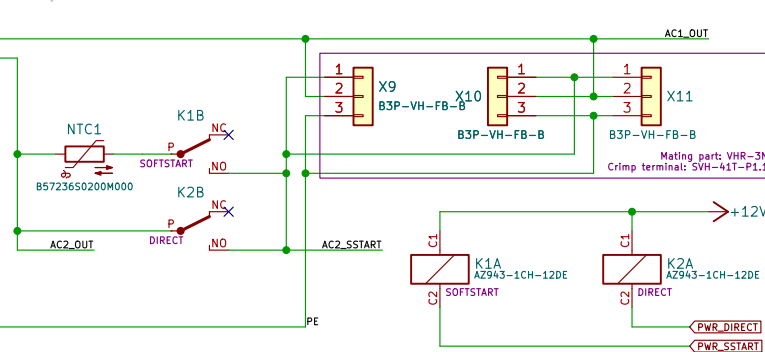
AC inlet, filtering and protection



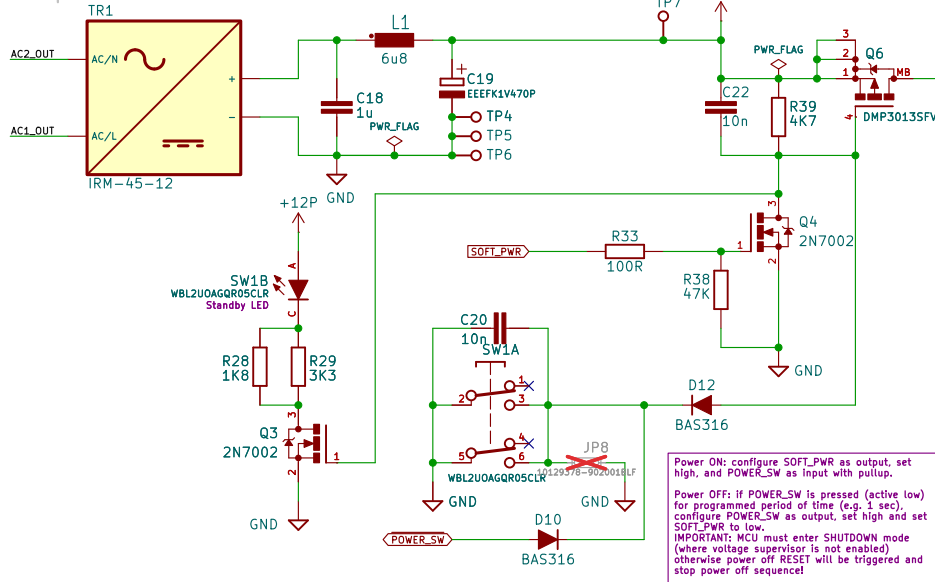
EMI compliance filter



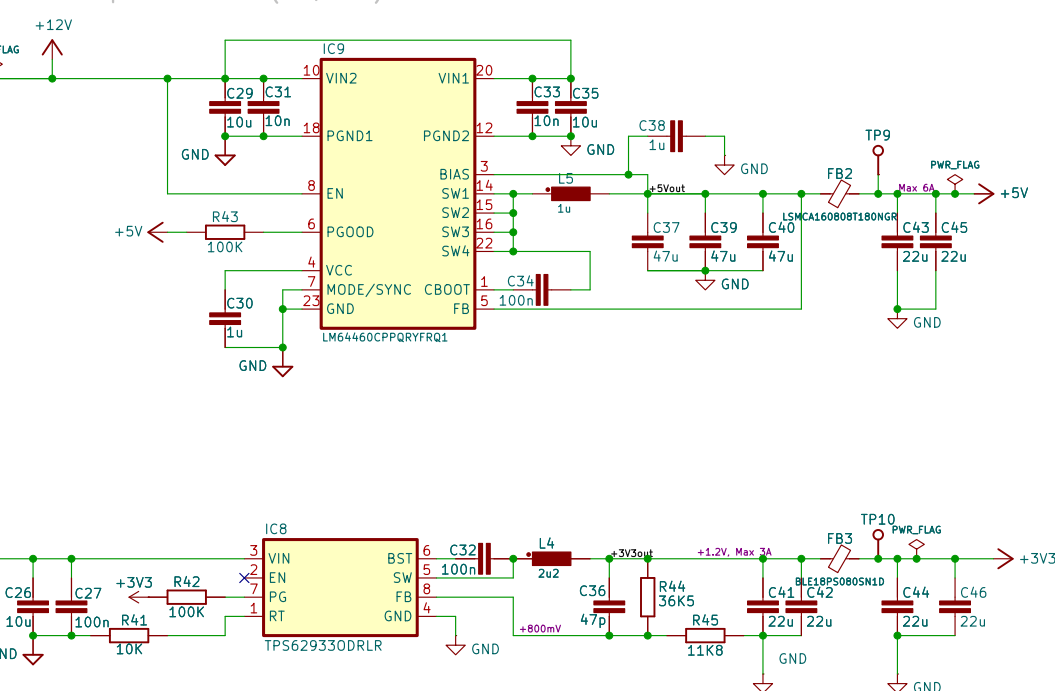
AC power control, Inrush current limiter



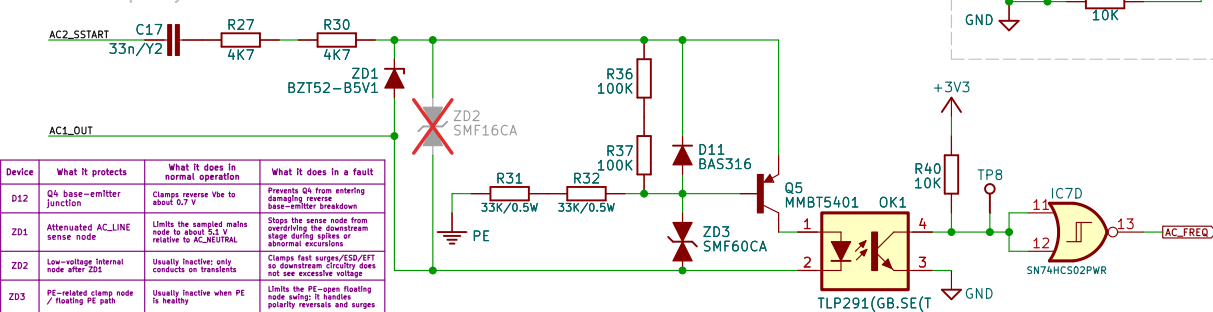
Soft power switch



DC-DC stepdown converters (+5V, +3V3)



AC Mains frequency and PE fault detection



Device	What it protects	What it does in normal operation	What it does in a fault
D12	Q4 base-emitter junction	Clamps reverse Vbe to about 0.7 V	Prevents Q4 from entering damaging reverse base-emitter breakdown
ZD1	Attenuated AC_LINE sense node	Limits the sampled mains node to about 5.1 V relative to AC_NEUTRAL	Stops the sense node from overvoltage during spikes or abnormal excursions
ZD2	Low-voltage internal node after ZD1	Usually inactive; only conducts on transients	Clamps fast surges/ESD/RFI so downstream circuitry does not see excessive voltage
ZD3	PE-related clamp node / floating PE path	Usually inactive when PE is healthy	Limits the PE-open floating node swing; it handles polarity reversals and surges

<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: /AC power, DC power, AC sync/

File: ac_dc_power.kicad_sch

Title: BB3+ mainboard

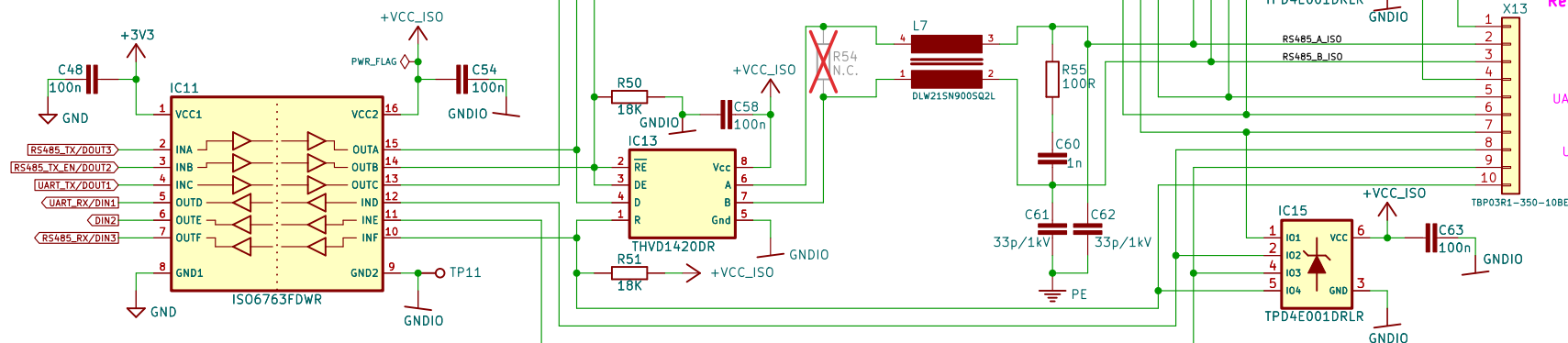
Size: A4 Date: 2026-10-05

KiCad E.D.A. 10.0.6

Rev: r1B3

Id: 4/10

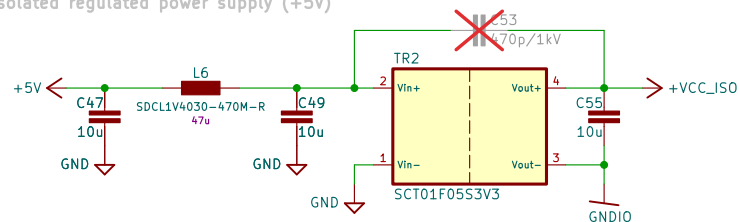
Isolated digital I/O & RS485 transceiver (half-duplex)



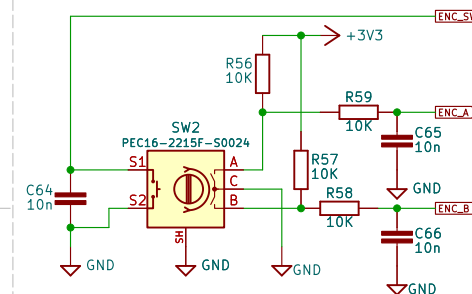
Rear panel I/O connector

- +VCC_ISO 1
- RS485_A_ISO 2
- RS485_B_ISO 3
- GND_ISO 4
- UART_TX/DOUT1 5
- DOUT2 6
- DOUT3 7
- UART_RX/DIN1 8
- DIN2 9
- DIN3 10

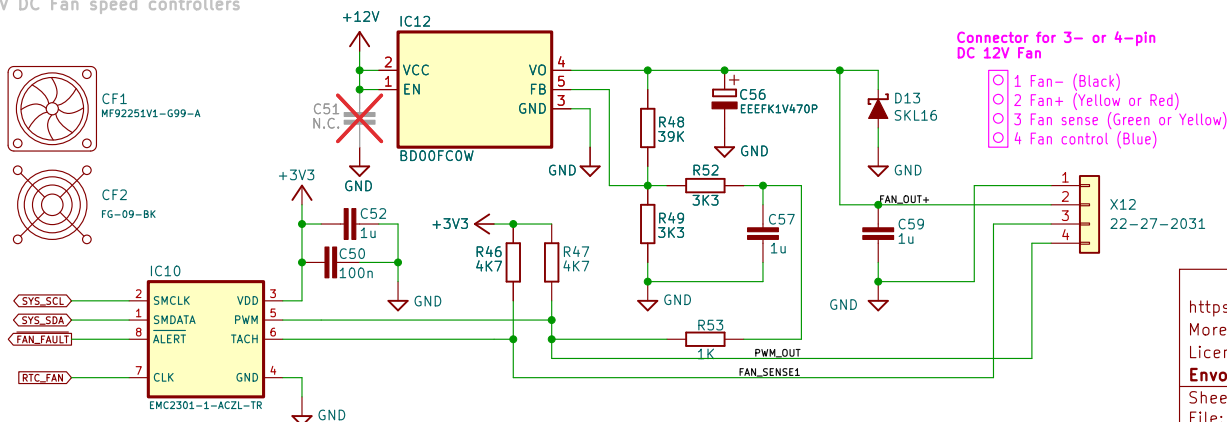
Isolated regulated power supply (+5V)



Encoder with switch



12V DC Fan speed controllers



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: /Isolated digital IO, Fan controller, Encoder/

File: digital_io_fan_encoder.kicad_sch

Title: BB3+ mainboard

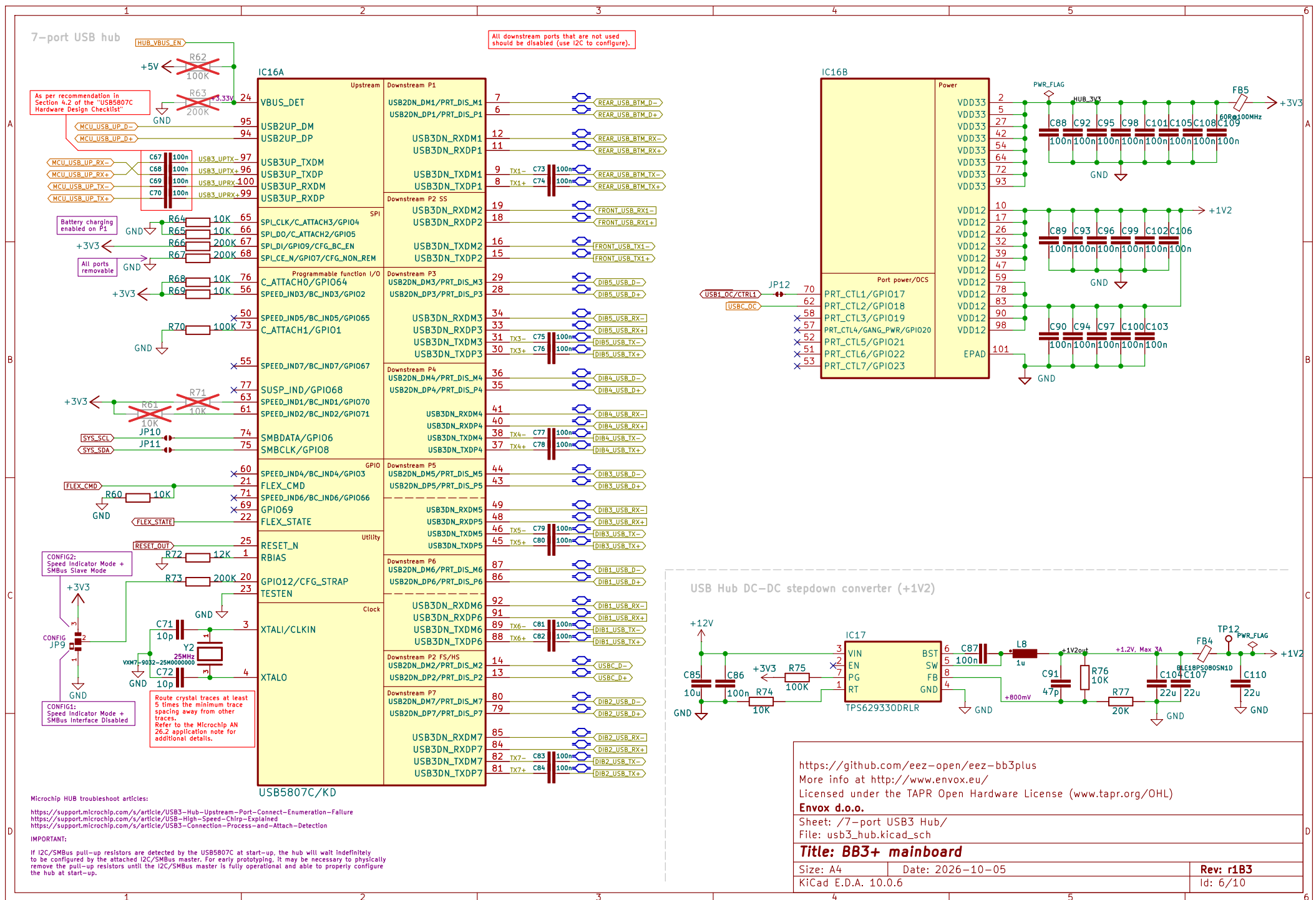
Size: A4

Date: 2026-10-05

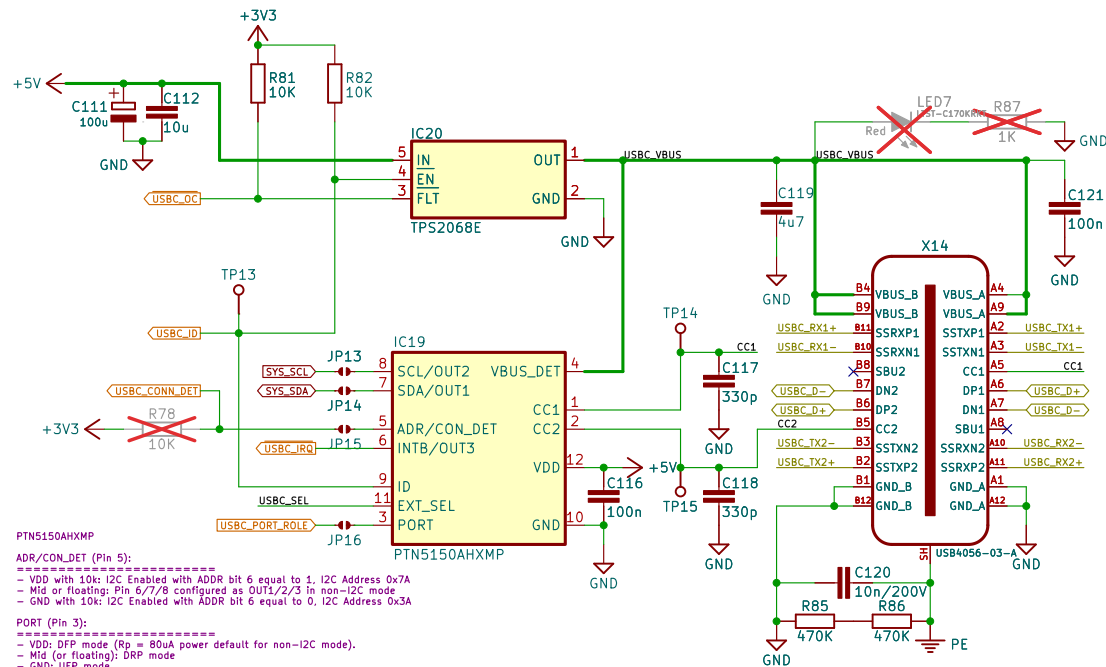
KiCad E.D.A. 10.0.6

Rev: r1B3

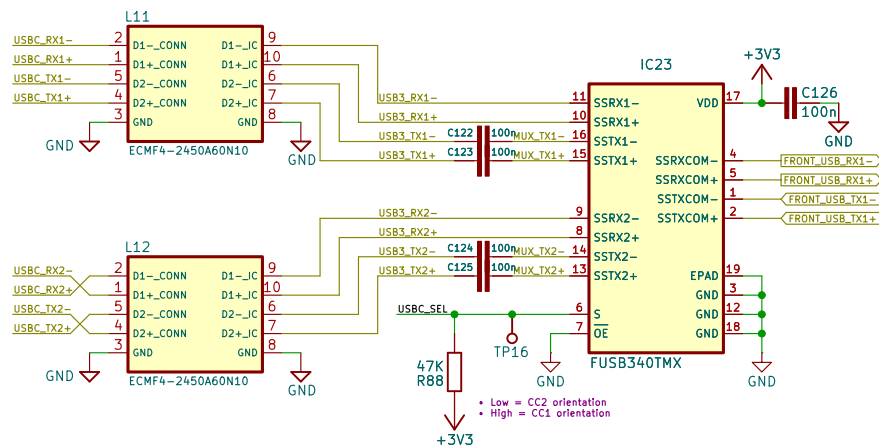
Id: 5/10



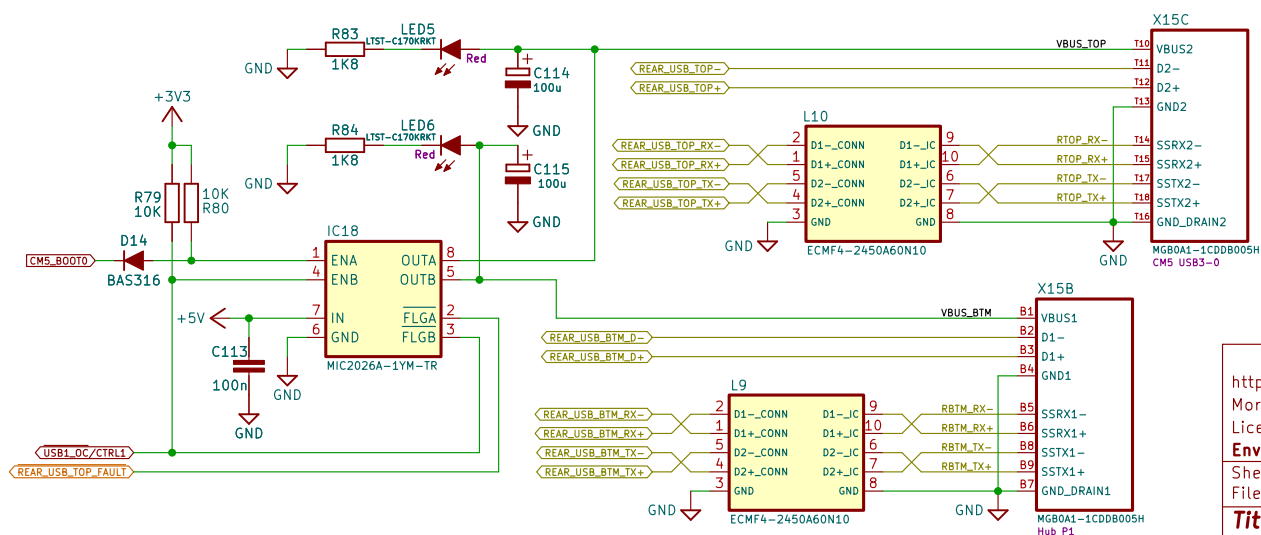
USB-C 3.0 front port (source max 1.5A)



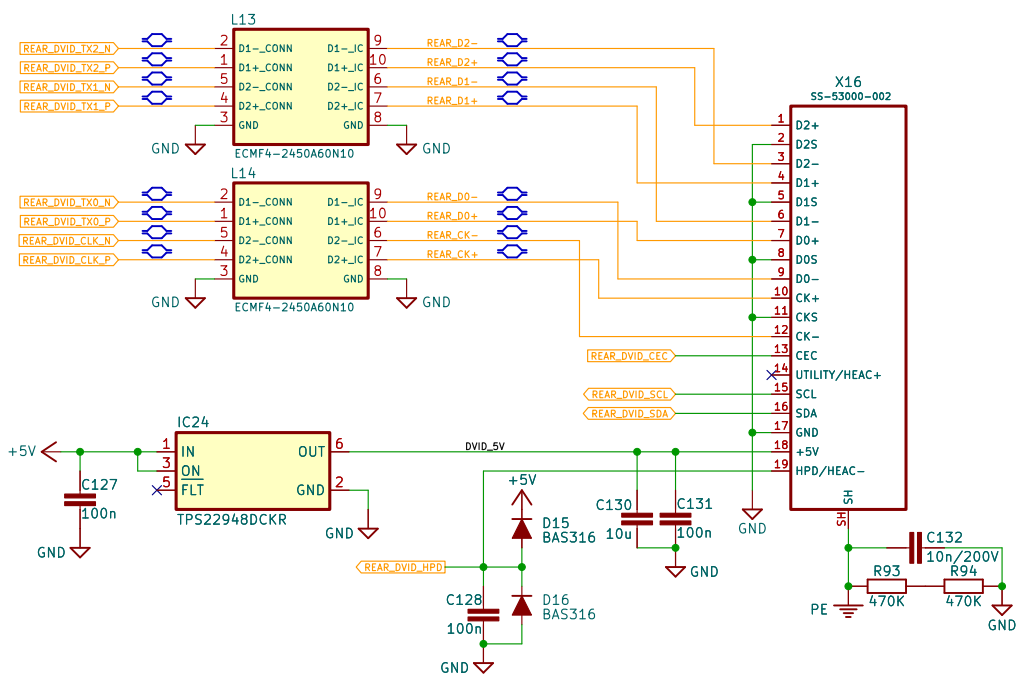
USB-C 3.0 Protection and Mux



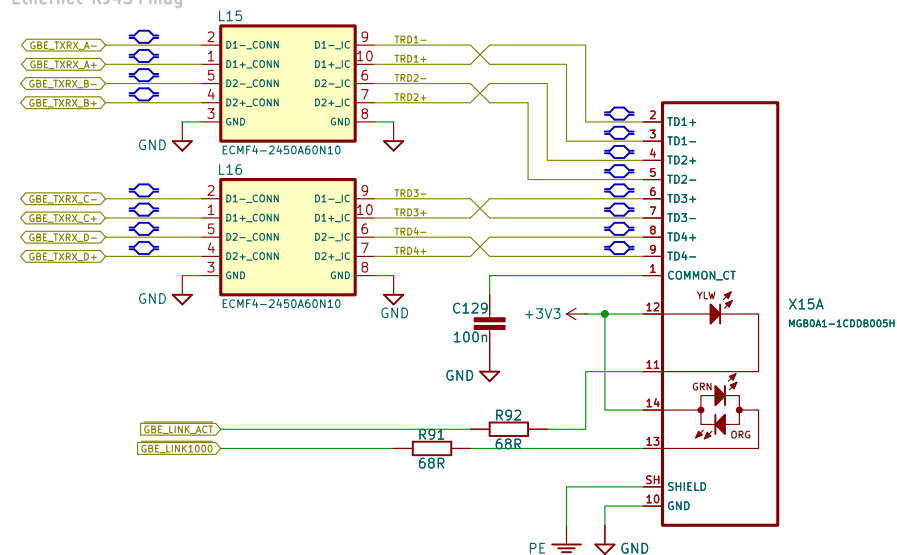
USB2.0 Type-A (Rear panel)



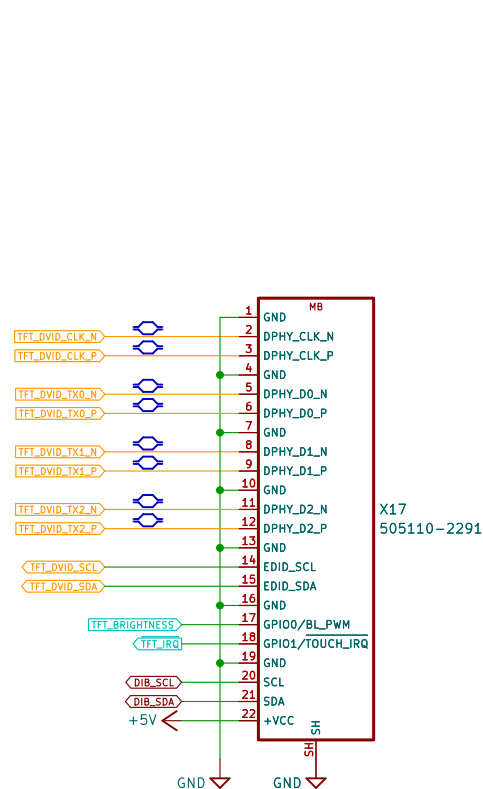
External monitor digital output



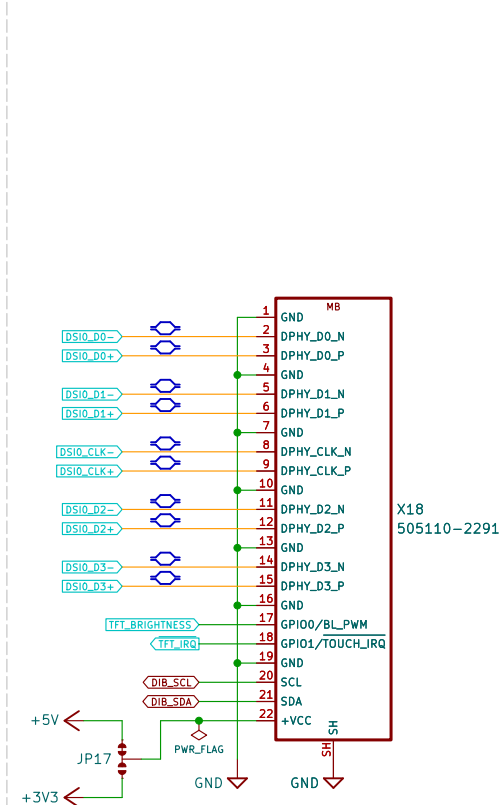
Ethernet RJ45+mag



TFT Touchscreen display DVID output (22-pin FFC)



TFT Touchscreen display DSI output (22-pin FFC)



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: /Ethernet, Video outputs/

File: video_outputs.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-10-05

KiCad E.D.A. 10.0.6

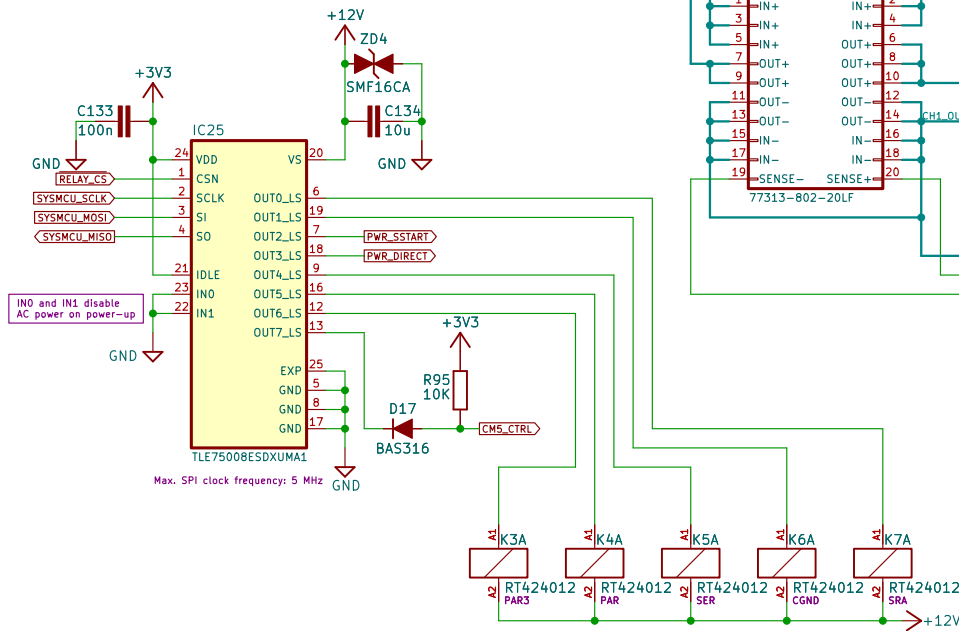
Rev: r1B3

Id: 8/10

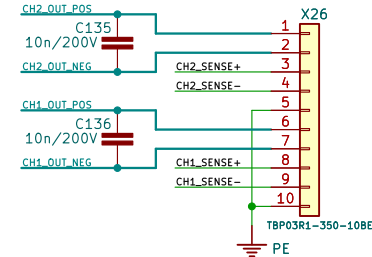
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

PWR+ IN 1	2 PWR+ IN
PWR+ IN 3	4 PWR+ IN
PWR+ IN 5	6 PWR+ OUT
PWR+ OUT 7	8 PWR+ OUT
PWR+ OUT 9	10 PWR+ OUT
PWR- OUT 11	12 PWR- OUT
PWR- OUT 13	14 PWR- OUT
PWR- IN 15	16 PWR- IN
PWR- IN 17	18 PWR- IN
SENSE- 19	20 SENSE+



Rear front power outputs



Rear panel power connector

CH2_OUT+ 1	2
CH2_OUT- 3	4
CH2_SENSE+ 5	6
CH2_SENSE- 7	8
PE 9	10
CH1_OUT+ 11	12
CH1_OUT- 13	14
CH1_SENSE+ 15	16
CH1_SENSE- 17	18
PE 19	20

<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: //Power modules coupling/

File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-10-05

KiCad E.D.A. 10.0.6

Rev: r1B3

Id: 10/10